

PITAR: an LLM-powered Agent towards Intelligent and Accurate Manipulations in Extended Reality with Multimodal Interactions

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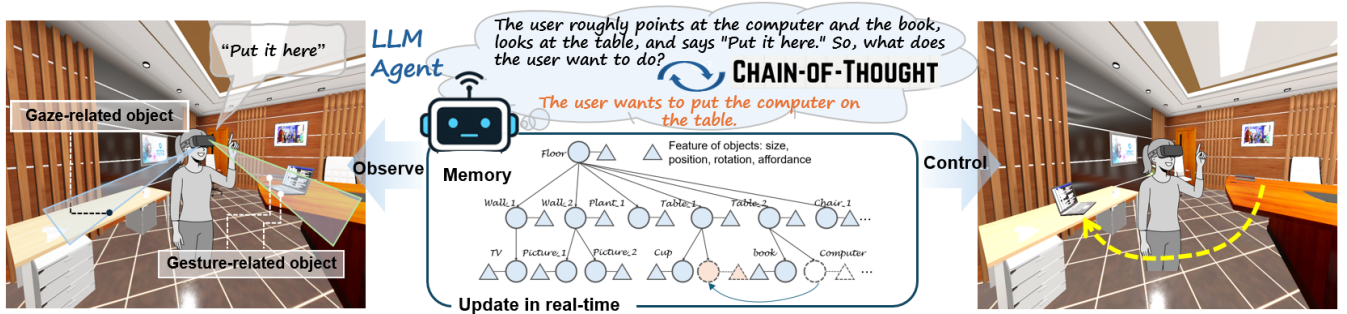


Figure 1: PITAR enables natural interaction in XR through multimodal inputs. It maintains a dynamic scene memory and infer user intent, which is then translated into real-time low-level commands for context-aware object manipulation.

Abstract

We present PITAR, an large language model (LLM)-powered agent for intelligent and accurate manipulations in extended reality (XR) through multimodal interactions. PITAR integrates eye gaze, pointing gesture, and speech—particularly pronoun-based commands—to correctly infer user intent and control virtual objects. Using real-time data from XR headset and a few-shot prompting strategy, PITAR performs joint reasoning over multimodal signals and memory of the scenario to identify the target object and determine desired action and interaction parameters. A prototype VR system implementing PITAR demonstrates intuitive, human-like communication between users and virtual environments, advancing the development of intelligent agents for immersive interaction.

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CCS Concepts

• Human-centered computing → Virtual reality; Natural language interfaces; Interaction techniques.

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1 Introduction

People often use pronouns like “put it here” in daily communication and easily resolve such ambiguities. As intelligent systems become more embedded in everyday life, interpreting underspecified expressions has become a central challenge in human-computer interaction. With the growing integration of XR and AI, multimodal interaction systems that combine gaze and gesture have attracted increasing attention. Yet most rely on a single modality—such as gaze or pointing—and struggle with phrases like “put it over there.” They also assume precise pointing [Bolt 1980], neglecting the inherent ambiguity of reference in real interactions. As a result, systems

often fail when gaze or gesture cues are unclear. Moreover, existing work focuses largely on question answering [Lee et al. 2024], offering limited support for direct manipulation of virtual objects in immersive environments.

We propose PITAR, a LLM-powered agent for natural and intelligent control in XR (Fig. 1). PITAR jointly reasons over gaze, gesture, and language to interpret and execute complex commands, including those with pronouns. It infers the intended target object, action, and manipulation parameters such as position, scale, or orientation. We also developed a VR prototype and conducted a preliminary user study, which showed that PITAR enables intuitive, human-like communication in virtual environments, advancing intelligent agents for immersive interaction.

2 Method

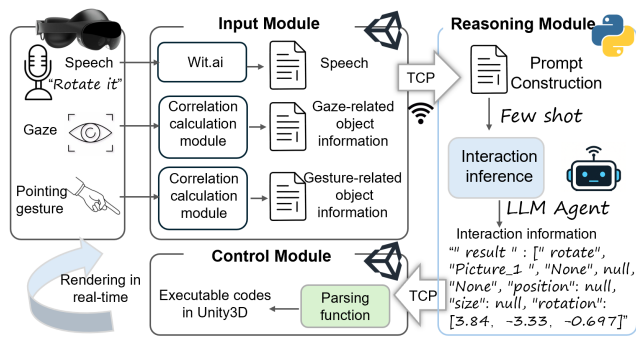


Figure 2: Structure overview of PITAR.

2.1 Structural Design of PITAR

PITAR is composed of three main components as shown in Fig. 2. The Input Module captures speech, gaze, and pointing gestures, with speech transcribed into text. This multimodal information is passed to the Reasoning Module, which constructs prompts from the text and contextual object information. An LLM then infers the intended target, action, and control parameters (e.g., position, orientation, scale). PITAR encodes all scene objects and attributes in a dynamic memory that updates in real time, organized as a tree graph where nodes represent objects and arrows represent their relations. Finally, the Control Module interprets the LLM output and maps it to Unity commands, enabling precise manipulation of virtual objects in the immersive environment.

2.2 Implementation

We developed a PITAR prototype in VR on a Meta Quest Pro headset using Unity3D. The headset captures gaze, gesture, and speech data, with speech transcribed in real time via Wit.ai¹. The app samples recent gaze and gesture candidates and computes a relevance score between rays and nearby objects, considering those within a 15° cone and using perpendicular distance from the object's center to the ray.

¹<https://wit.ai>

All multimodal data—including transcribed speech and gaze/gesture targets—are transmitted via TCP to the Reasoning Module (Intel Core Ultra 5, 64GB RAM, RTX 4090) running the LLM model (deepseek-r1 32b). The module constructs prompts and performs few-shot reasoning to infer user intent (object, action, parameters). Results are then sent back via TCP, where a parser maps the structured output to executable VR commands.

2.3 Experiments and Results

To evaluate PITAR, we conducted a preliminary study with 5 participants (3 male, 2 female, aged 22–30) using the VR prototype. None had XR experience and were asked to complete interactive tasks, including selecting, deleting, moving, and resizing objects with pronoun-based voice commands combined with gaze and pointing (Fig. 3). Participants reported the interaction as natural, intuitive, and engaging, especially appreciating the system's handling of vague expressions like “put it there” or “make it bigger.” These findings highlight the potential of LLM-based agents for context-aware, pronoun-resolving control in XR. A 3–5s delay was observed due to communication and inference; while acceptable for interaction, reducing latency is a priority for future work.



Figure 3: Examples in the experiment. (a) PITAR scales the TV up. (b) PITAR translates the cup to the user's hand.

3 Conclusion and Future Work

PITAR demonstrates how integrating gaze, gestures, and speech with LLM reasoning enables intuitive and context-aware interaction in XR. It offers a step toward more human-like communication with intelligent agents in immersive settings.

Acknowledgments

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